PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 05 – Static Members and Object Lifecycle

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Task05.md

Aim

To demonstrate static variables, static methods and the object lifecycle in the Campus Smart-Parking System.

Static Members

Static members belong to the class rather than individual objects.

- **Static variable:** Shared by all objects.
- **Static method:** Can be called using the class name.
- **Instance variable:** Stores separate data for each object.

Object Lifecycle

1. An object is created using the `new` keyword.
2. Memory is allocated and the constructor initializes the object.
3. The object is used through its reference.
4. When an object is no longer reachable, it becomes eligible for garbage collection.

Application

The parking system uses a static variable to count the total number of tickets created. This avoids maintaining a separate ticket count in every object.

Conclusion

Static members help maintain shared information, while object lifecycle management helps understand memory usage in Java.

ParkingTicket.java

```
public class ParkingTicket {
```

```
// Static variable shared by all objects
static int totalTickets = 0;

// Instance variables
int ticketId;
String vehicleNumber;

// Constructor
ParkingTicket(int ticketId, String vehicleNumber) {
   this.ticketId = ticketId;
   this.vehicleNumber = vehicleNumber;
   totalTickets++;
}

// Instance method
void displayTicket() {
   System.out.println("Ticket ID: " + ticketId);
   System.out.println("Vehicle Number: " + vehicleNumber);
}

// Static method
static void displayTotalTickets() {
   System.out.println("Total Tickets: " + totalTickets);
}

public static void main(String[] args) {

   ParkingTicket t1 =
      new ParkingTicket(101, "TN01AB1234");

   ParkingTicket t2 =
      new ParkingTicket(102, "TN02CD5678");

   ParkingTicket t3 =
      new ParkingTicket(103, "TN03EF9012");

   System.out.println("Parking Ticket Details:");

   t1.displayTicket();
   System.out.println();

   t2.displayTicket();
   System.out.println();

   t3.displayTicket();
   System.out.println();

   ParkingTicket.displayTotalTickets();

   // Object reference is removed
```

```
        t1 = null;
    }
}
```

Output:

